

Impact of Weather on US Aviation Operations and Costs

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March 6, 2026

Project Description

This project explores how weather events reported by the NOAA impact flight performance, passenger volumes, and ticket costs in the USA. We aim to understand how specific storms or seasonal patterns change airline operations and influence quarterly fares, specifically answering: **How do severe weather events impact flight delays, passenger volume, and quarterly ticket fares across the United States?**

Datasets

We will use four main datasets to perform our analysis:

- **NOAA Daily Summaries:** This dataset provides local weather records from stations across the US. We will use it to track daily measurements of precipitation (rain/snow), temperature, and wind speeds;
- **BTS Flight Statistics:** Sourced from the Bureau of Transportation Statistics, this contains detailed records of domestic flights. It includes key metrics like scheduled vs. actual departure times, cancellation codes, and flight trajectories;
- **TSA Checkpoint Travel Numbers:** This provides the total daily count of passengers screened at US airports;
- **Average Domestic Airline Fares:** This quarterly dataset shows the average price paid for flights based on the city of origin..

Data Warehousing (DW)

- Does extreme weather significantly impact travel time, flight cancellations, or flight trajectories in specific geographical locations? (Diagnostic)
- Are flight delays related to the daily passenger volumes recorded at TSA checkpoints? (Diagnostic)
- Does the number of flights or the geographical location of a city influence the average ticket price? (Descriptive)
- How do ticket prices vary between "stormy" and "clear" quarters, and do high ticket prices influence the rate of cancellations? (Descriptive and Diagnostic)

Data Mining (DM)

- Based on historical weather patterns and city travel numbers, what are the predicted ticket prices for the next quarter? (Predictive)
- Which months and cities are recommended for travel to minimize the risk of weather-related delays and high costs? (Prescriptive)

Datasets Links

- NOAA: <https://www.ncei.noaa.gov/>
- BTS: <https://www.bts.gov/>
- TSA: <https://www.tsa.gov/>
- Fares: <https://data.transportation.gov/>